Setting up the Environment

- Download Flutter SDK
- Check your Flutter Installation
- Installation Setup for Visual Studio Code
- Installation Setup for Android Studio

Download Flutter SDK

1. **Download the Flutter SDK:**
    - Windows: https://flutter.dev/docs/get-started/install/windows
    - macOS: https://flutter.dev/docs/get-started/install/macos
2. **Extract the Flutter SDK:**
    - Extract the downloaded zip file and move the 'flutter' directory to your desired installation location for the Flutter SDK. For example: `C:\src\flutter`
3. **Update the System Path:**
    - Open the Environment Variables settings:

      Option 1: Right-click on My Computer or This PC, select Properties, then go to Advanced System Settings.

      Option 2: Simply type "env" in the Windows search bar and select Edit the system environment variables.
    - Under the Advanced tab, click Environment Variables.
    - In User Variables, find the Path variable and click Edit.
    - Click New and add the path to the flutter/bin directory. For example: `C:\src\flutter\bin`
    - Click OK to save changes.

Check your Flutter Installation

1. Open the Command Prompt, or simply type "cmd" in the Windows search bar. Navigate to the Flutter directory's bin folder by typing

   ```
   cd C:\src\flutter\bin
   ```
2. Run the flutter doctor by typing:

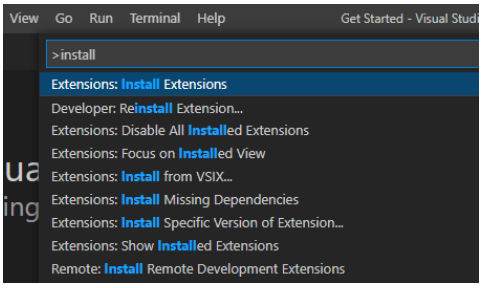
   ```
   C:\src\flutter\bin> flutter doctor
   ```

3. Example output:

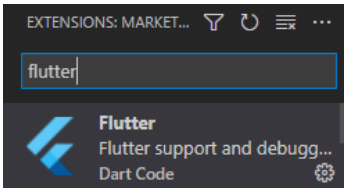


Installation Setup for Visual Studio Code

1. **Download and install the Visual Studio Code**
    - Visit https://code.visualstudio.com and download the installer for your operating system.
    - Follow the on-screen instructions to install Visual Studio Code
2. **Launch the Visual Studio Code**
    - Open the Visual Studio Code IDE after installation.
3. **Install Extensions**
    - Option 1: Navigate to the View tab and select Command Palette (or press Ctrl + Shift + P).
    - Type "install" in the search bar and choose Extensions: Install Extensions from the dropdown.

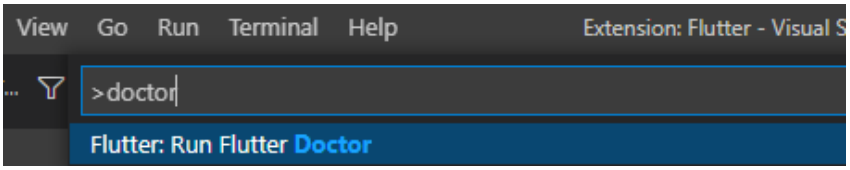


4. **Add the Flutter Extension**
    - In the Marketplace, type Flutter in the search bar.



    - Select the Flutter extension from the results and click the Install button.

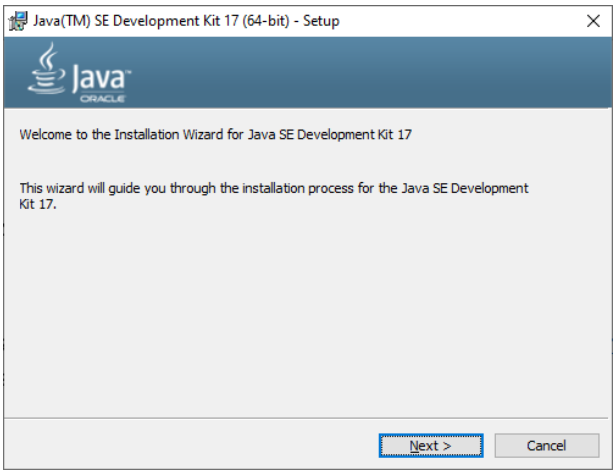
5. **Verify Flutter Setup**
    - Open the terminal within Visual Studio Code or access the Command Palette (Ctrl + Shift + P).
    - Run flutter doctor to check the status of your Flutter setup.



    - Review the output for any required actions or missing components.

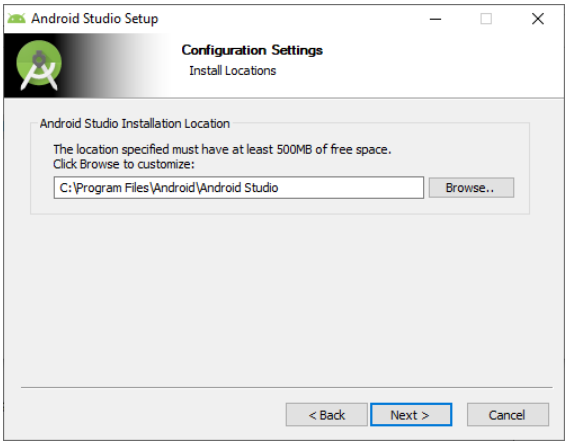
Installation Setup for Android Studio

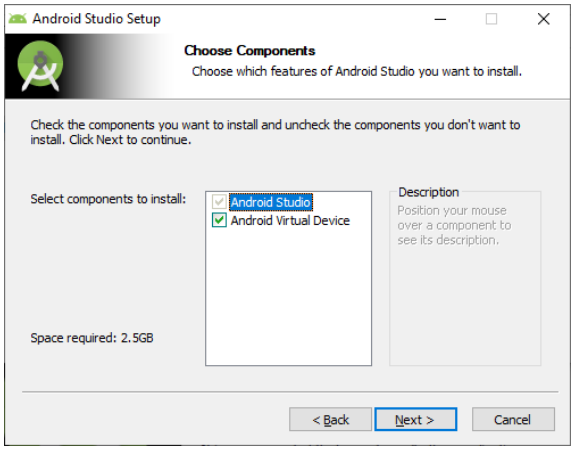
1. **Download and Install Java SDK**
    - Visit the Oracle Java SDK Download page: https://www.oracle.com/java/technologies/downloads/
    - Download the appropriate version of the Java SDK for your operating system.
    - Install the Java SDK using the default settings and complete the setup.

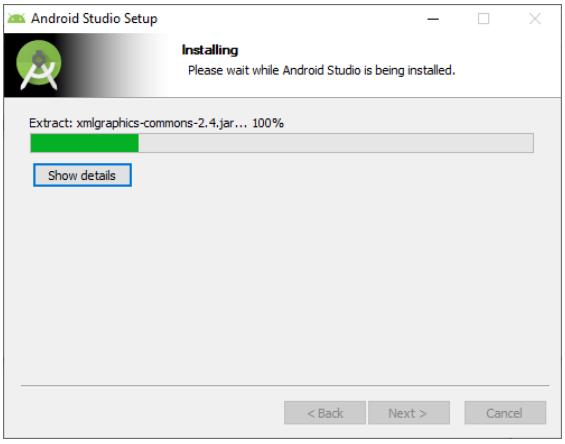


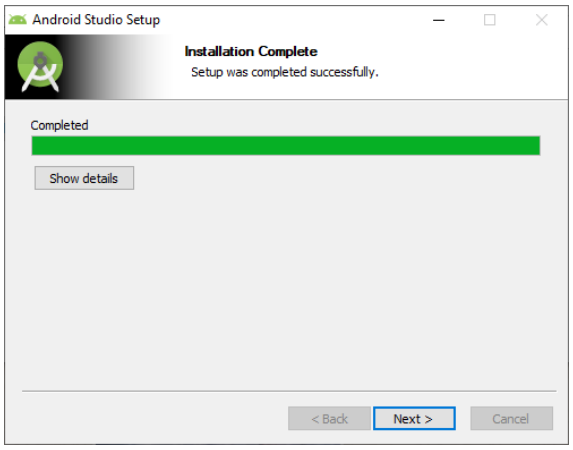
2. **Download and Install Android Studio**
    - Visit the Android Studio download page: https://developer.android.com/studio
    - Download the installer for your operating system and run it.
    - Follow the setup wizard and install Android Studio using the default settings.

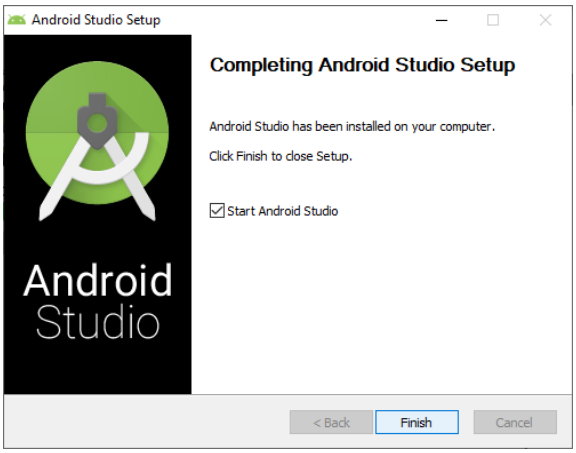

Android Studio Setup
Welcome to Android Studio Setup
Setup will guide you through the installation of Android Studio.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Android Studio
< Back
Next >
Cancel


Android Studio Setup
Configuration Settings
Install Locations
Android Studio Installation Location
The location specified must have at least 500MB of free space.
Click Browse to customize:
C:\Program Files\Android\Android Studio
Browse..
< Back
Next >
Cancel

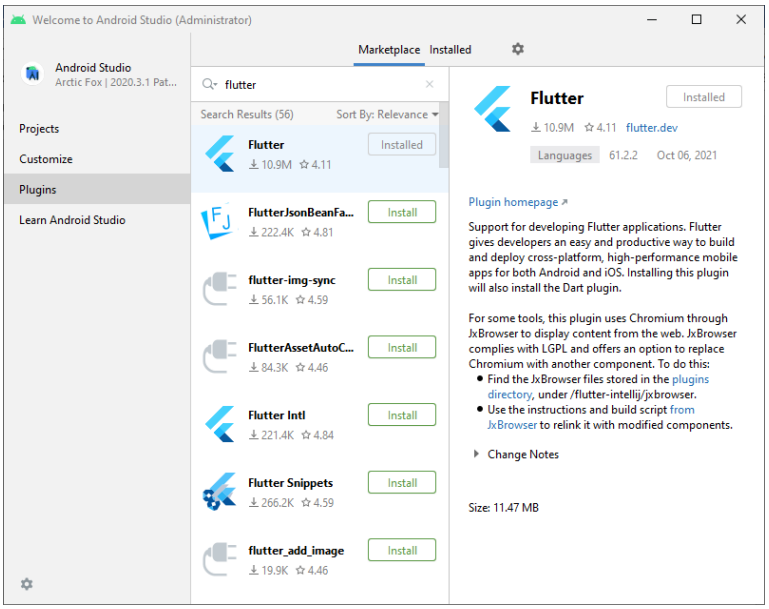

Android Studio Setup
Choose Components
Choose which features of Android Studio you want to install.
Check the components you want to install and uncheck the components you don't want to install. Click Next to continue.
Select components to install:
Android Studio
Android Virtual Device
Description
Position your mouse over a component to see its description.
Space required: 2.5GB
< Back
Next >
Cancel


Android Studio Setup
Installing
Please wait while Android Studio is being installed.
Extract: xmlgraphics-commons-2.4.jar... 100%
Show details
< Back
Next >
Cancel


Android Studio Setup
Installation Complete
Setup was completed successfully.
Completed
Show details
< Back
Next >
Cancel


Android Studio Setup
Completing Android Studio Setup
Android Studio has been installed on your computer.
Click Finish to close Setup.
Start Android Studio
Android Studio
< Back
Finish
Cancel

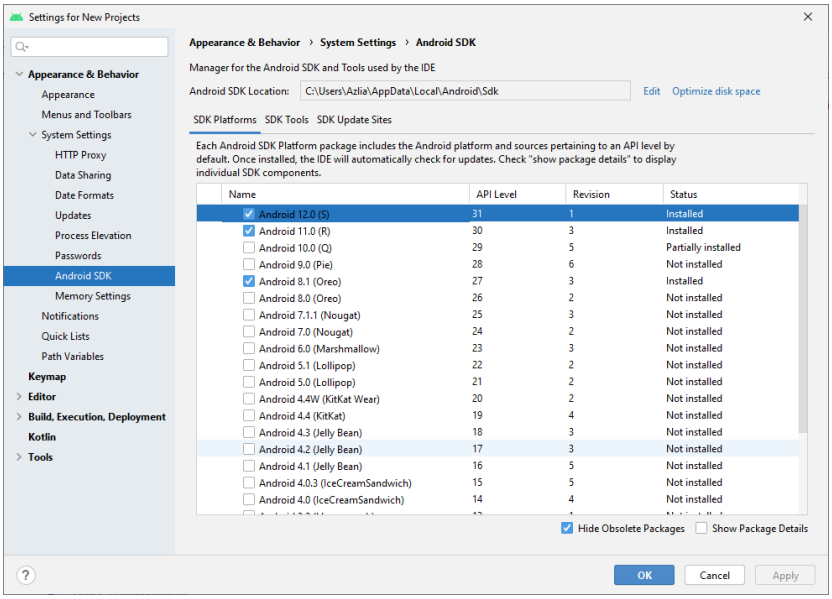
3. **Configure Android Studio for Flutter Development**
    - Launch Android Studio after installation.
    - Go to File > Settings > Plugins (or Preferences on macOS).
    - Search for Flutter in the plugins marketplace.



    - Install the Flutter plugin (it will also prompt you to install the Dart plugin if not already installed).
    - Restart Android Studio to apply changes.
4. **Verify Setup**
    - Open a terminal or Command Prompt.
    - Run flutter doctor to ensure Android Studio and its components are correctly set up.
    - Follow any additional instructions or install missing dependencies as prompted by flutter doctor.

Install Additional Android SDK

1. **Open the Android SDK Manager:**
    - Launch Android Studio.
    - Navigate to File > Settings > Appearance & Behavior > System Settings > Android SDK (on macOS, go to Android Studio > Preferences).
    - The Android SDK Manager window will open.



2. Install Required SDK Packages:

- In the SDK Platforms tab, select the required Android versions. It is recommended to install the latest version for Flutter development.
- Switch to the SDK Tools tab, and ensure the following tools are installed:

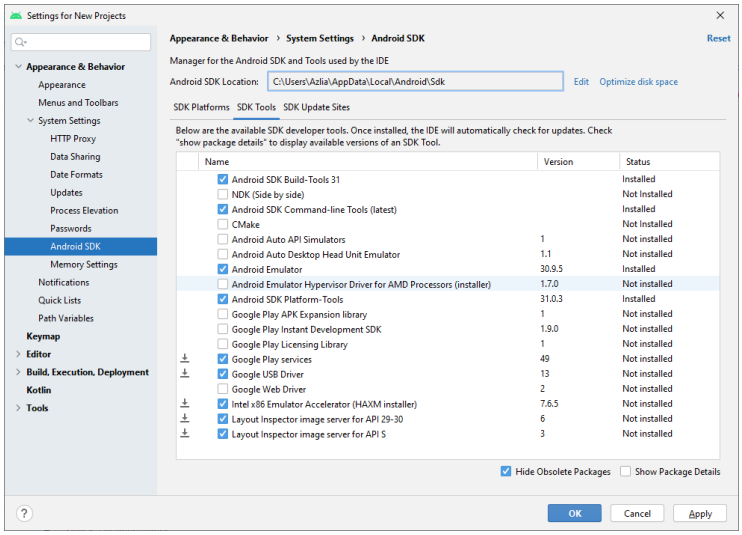
  Android SDK Build-Tools

  Android Emulator

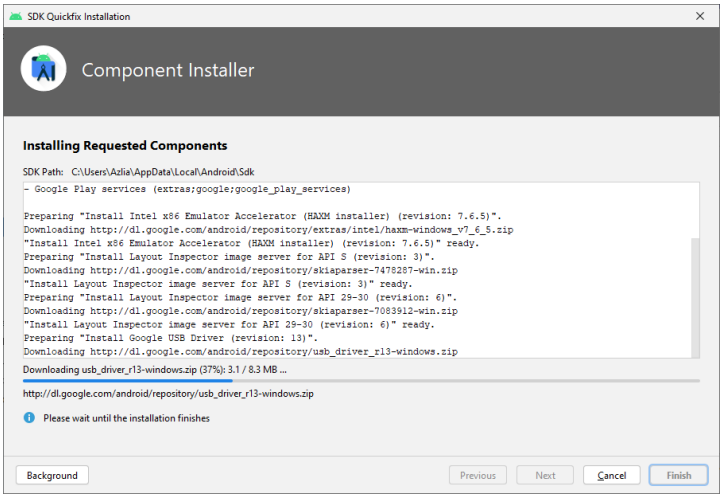
  Android SDK Platform-Tools

  Android SDK Tools (if available)

  Google Play Services (if needed)



3. **Install Missing Packages:**
    - If Flutter prompts any missing SDK components during flutter doctor, return to the Android SDK Manager and install the required packages.
4. **Apply Changes:**
    - Click Apply or OK to download and install the selected packages.



5. **Verify Installation:**
    - Open a terminal or Command Prompt.
    - Run flutter doctor to confirm that the Android SDK is correctly configured and there are no missing components.